codata

Table of Contents

codata 1
codata_cli 40
codata_constant_type 41
codata_version 42

NAME

codata - libray for fundamental physical constants

LIBRARY

codata (**-libcodata**, **-lcodata**)

SYNOPSIS

```
Fortran: use codata

C: include "codata.h"

Python: import pycodata
```

DESCRIPTION

codata(3) is a Fortran library providing the fundamental physical constants published by the CODATA.

Fortran API
exposes the constants as derived type codata_constant_type which carries:

- name
- value
- uncertainty
- unit

C API exposes a structure codata_constant_type that defines the same members as in Fortran:

- name
- value
- uncertainty
- unit

Python exposes a dictionary with the keys being:

- the name
- the value
- the uncertainty
- the unit

NOTES

The latest values (2022) do not have the year as a suffix in their name whereas older values (2010, 2014, 2018) feature the year as a suffix. All constants are declared with the same variable name in Fortran, C, and Python.

EXAMPLE

Example in Fortran

```
program example_in_f
use codata
implicit none
print '(A)', '########## EXAMPLE IN FORTRAN ##########'
print '(A)', '# VERSION'
print *, "version = ", version()
print '(A)', '# CONSTANTS'
print *, "c = ",  SPEED_OF_LIGHT_IN_VACUUM%value
print '(A)', '# UNCERTAINTY'
print *, "u(c) = ", SPEED_OF_LIGHT_IN_VACUUM%uncertainty
print '(A)', '# OLDER VALUES'
```

```
print '(A, F23.16)', "Mu_2022(latest) = ", MOLAR_MASS_CONSTANT%value
print '(A, F23.16)', "Mu_2018 = ", MOLAR_MASS_CONSTANT_2018%value
print '(A, F23.16)', "Mu_2014 = ",  MOLAR_MASS_CONSTANT_2014%value
print '(A, F23.16)', "Mu_2010 = ",  MOLAR_MASS_CONSTANT_2010%value
end program
```

Example in C

```
#include <stdio.h>
#include "codata.h"


int main(void){
printf("%f", c);
printf("########## EXAMPLE IN C ##########0);
printf("%s0,"# VERSION");
printf("version = %s0, codata_version());
printf("%s0,"# CONSTANTS");
printf("c = %f0, SPEED_OF_LIGHT_IN_VACUUM.value);
printf("%s0,"# UNCERTAINTY");
printf("u(c) = %f0, SPEED_OF_LIGHT_IN_VACUUM.uncertainty);
printf("%s0,"# OLDER VALUES");
printf("Mu_2022(latest) = %23.16f0, MOLAR_MASS_CONSTANT.value);
printf("Mu_2018 = %23.16f0, MOLAR_MASS_CONSTANT_2018.value);
printf("Mu_2014 = %23.16f0, MOLAR_MASS_CONSTANT_2014.value);
printf("Mu_2010 = %23.16f0, MOLAR_MASS_CONSTANT_2010.value);
return 0;
}
```

Example in Python

```
import sys
sys.path.insert(0, "../py/src/")
import pycodata
print("########## EXAMPLE IN PYTHON ##########")
print("# VERSION")
print(f"version = {pycodata.__version__}")
print("# Constants")
print("c =", pycodata.SPEED_OF_LIGHT_IN_VACUUM["value"])
print("# UNCERTAINTY")
print("u(c) = ", pycodata.SPEED_OF_LIGHT_IN_VACUUM["uncertainty"])
print("# OLDER VALUES")
print("Mu_2022 = ", pycodata.MOLAR_MASS_CONSTANT["value"])
print("Mu_2018 = ", pycodata.MOLAR_MASS_CONSTANT_2018["value"])
print("Mu_2014 = ", pycodata.MOLAR_MASS_CONSTANT_2014["value"])
print("Mu_2010 = ", pycodata.MOLAR_MASS_CONSTANT_2010["value"])
```

SEE ALSO

codata(1), **codata_version**(3), **codata_constant_type**(3)

CODATA 2022

List of available constants:

- ALPHA_PARTICLE_ELECTRON_MASS_RATIO
- ALPHA_PARTICLE_MASS
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_IN_MEV
- ALPHA_PARTICLE_MASS_IN_U
- ALPHA_PARTICLE_MOLAR_MASS
- ALPHA_PARTICLE_PROTON_MASS_RATIO
- ALPHA_PARTICLE_RELATIVE_ATOMIC_MASS
- ALPHA_PARTICLE_RMS_CHARGE_RADIUS
- ANGSTROM_STAR
- ATOMIC_MASS_CONSTANT
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_IN_MEV
- ATOMIC_MASS_UNIT_ELECTRON_VOLT_RELATIONSHIP
- ATOMIC_MASS_UNIT_HARTREE_RELATIONSHIP
- ATOMIC_MASS_UNIT_HERTZ_RELATIONSHIP
- ATOMIC_MASS_UNIT_INVERSE_METER_RELATIONSHIP
- ATOMIC_MASS_UNIT_JOULE_RELATIONSHIP
- ATOMIC_MASS_UNIT_KELVIN_RELATIONSHIP
- ATOMIC_MASS_UNIT_KILOGRAM_RELATIONSHIP
- ATOMIC_UNIT_OF_1ST_HYPERPOLARIZABILITY
- ATOMIC_UNIT_OF_2ND_HYPERPOLARIZABILITY
- ATOMIC_UNIT_OF_ACTION
- ATOMIC_UNIT_OF_CHARGE
- ATOMIC_UNIT_OF_CHARGE_DENSITY
- ATOMIC_UNIT_OF_CURRENT
- ATOMIC_UNIT_OF_ELECTRIC_DIPOLE_MOM
- ATOMIC_UNIT_OF_ELECTRIC_FIELD
- ATOMIC_UNIT_OF_ELECTRIC_FIELD_GRADIENT
- ATOMIC_UNIT_OF_ELECTRIC_POLARIZABILITY
- ATOMIC_UNIT_OF_ELECTRIC_POTENTIAL
- ATOMIC_UNIT_OF_ELECTRIC_QUADRUPOLE_MOM
- ATOMIC_UNIT_OF_ENERGY
- ATOMIC_UNIT_OF_FORCE
- ATOMIC_UNIT_OF_LENGTH
- ATOMIC_UNIT_OF_MAG_DIPOLE_MOM
- ATOMIC_UNIT_OF_MAG_FLUX_DENSITY
- ATOMIC_UNIT_OF_MAGNETIZABILITY

- ATOMIC_UNIT_OF_MASS
- ATOMIC_UNIT_OF_MOMENTUM
- ATOMIC_UNIT_OF_PERMITTIVITY
- ATOMIC_UNIT_OF_TIME
- ATOMIC_UNIT_OF_VELOCITY
- AVOGADRO_CONSTANT
- BOHR_MAGNETON
- BOHR_MAGNETON_IN_EV_T
- BOHR_MAGNETON_IN_HZ_T
- BOHR_MAGNETON_IN_INVERSE_METER_PER_TESLA
- BOHR_MAGNETON_IN_K_T
- BOHR_RADIUS
- BOLTZMANN_CONSTANT
- BOLTZMANN_CONSTANT_IN_EV_K
- BOLTZMANN_CONSTANT_IN_HZ_K
- BOLTZMANN_CONSTANT_IN_INVERSE_METER_PER_KELVIN
- CHARACTERISTIC_IMPEDANCE_OF_VACUUM
- CLASSICAL_ELECTRON_RADIUS
- COMPTON_WAVELENGTH
- CONDUCTANCE_QUANTUM
- CONVENTIONAL_VALUE_OF_AMPERE_90
- CONVENTIONAL_VALUE_OF_COULOMB_90
- CONVENTIONAL_VALUE_OF_FARAD_90
- CONVENTIONAL_VALUE_OF_HENRY_90
- CONVENTIONAL_VALUE_OF_JOSEPHSON_CONSTANT
- CONVENTIONAL_VALUE_OF_OHM_90
- CONVENTIONAL_VALUE_OF_VOLT_90
- CONVENTIONAL_VALUE_OF_VON_KLITZING_CONSTANT
- CONVENTIONAL_VALUE_OF_WATT_90
- COPPER_X_UNIT
- DEUTERON_ELECTRON_MAG_MOM_RATIO
- DEUTERON_ELECTRON_MASS_RATIO
- DEUTERON_G_FACTOR
- DEUTERON_MAG_MOM
- DEUTERON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- DEUTERON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- DEUTERON_MASS
- DEUTERON_MASS_ENERGY_EQUIVALENT

- DEUTERON_MASS_ENERGY_EQUIVALENT_IN_MEV
- DEUTERON_MASS_IN_U
- DEUTERON_MOLAR_MASS
- DEUTERON_NEUTRON_MAG_MOM_RATIO
- DEUTERON_PROTON_MAG_MOM_RATIO
- DEUTERON_PROTON_MASS_RATIO
- DEUTERON_RELATIVE_ATOMIC_MASS
- DEUTERON_RMS_CHARGE_RADIUS
- ELECTRON_CHARGE_TO_MASS_QUOTIENT
- ELECTRON_DEUTERON_MAG_MOM_RATIO
- ELECTRON_DEUTERON_MASS_RATIO
- ELECTRON_G_FACTOR
- ELECTRON_GYROMAG_RATIO
- ELECTRON_GYROMAG_RATIO_IN_MHZ_T
- ELECTRON_HELION_MASS_RATIO
- ELECTRON_MAG_MOM
- ELECTRON_MAG_MOM_ANOMALY
- ELECTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- ELECTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- ELECTRON_MASS
- ELECTRON_MASS_ENERGY_EQUIVALENT
- ELECTRON_MASS_ENERGY_EQUIVALENT_IN_MEV
- ELECTRON_MASS_IN_U
- ELECTRON_MOLAR_MASS
- ELECTRON_MUON_MAG_MOM_RATIO
- ELECTRON_MUON_MASS_RATIO
- ELECTRON_NEUTRON_MAG_MOM_RATIO
- ELECTRON_NEUTRON_MASS_RATIO
- ELECTRON_PROTON_MAG_MOM_RATIO
- ELECTRON_PROTON_MASS_RATIO
- ELECTRON_RELATIVE_ATOMIC_MASS
- ELECTRON_TAU_MASS_RATIO
- ELECTRON_TO_ALPHA_PARTICLE_MASS_RATIO
- ELECTRON_TO_SHIELDED_HELION_MAG_MOM_RATIO
- ELECTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO
- ELECTRON_TRITON_MASS_RATIO
- ELECTRON_VOLT
- ELECTRON_VOLT_ATOMIC_MASS_UNIT_RELATIONSHIP

- ELECTRON_VOLT_HARTREE_RELATIONSHIP
- ELECTRON_VOLT_HERTZ_RELATIONSHIP
- ELECTRON_VOLT_INVERSE_METER_RELATIONSHIP
- ELECTRON_VOLT_JOULE_RELATIONSHIP
- ELECTRON_VOLT_KELVIN_RELATIONSHIP
- ELECTRON_VOLT_KILOGRAM_RELATIONSHIP
- ELEMENTARY_CHARGE
- ELEMENTARY_CHARGE_OVER_H_BAR
- FARADAY_CONSTANT
- FERMI_COUPLING_CONSTANT
- FINE_STRUCTURE_CONSTANT
- FIRST_RADIATION_CONSTANT
- FIRST_RADIATION_CONSTANT_FOR_SPECTRAL_RADIANCE
- HARTREE_ATOMIC_MASS_UNIT_RELATIONSHIP
- HARTREE_ELECTRON_VOLT_RELATIONSHIP
- HARTREE_ENERGY
- HARTREE_ENERGY_IN_EV
- HARTREE_HERTZ_RELATIONSHIP
- HARTREE_INVERSE_METER_RELATIONSHIP
- HARTREE_JOULE_RELATIONSHIP
- HARTREE_KELVIN_RELATIONSHIP
- HARTREE_KILOGRAM_RELATIONSHIP
- HELION_ELECTRON_MASS_RATIO
- HELION_G_FACTOR
- HELION_MAG_MOM
- HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- HELION_MASS
- HELION_MASS_ENERGY_EQUIVALENT
- HELION_MASS_ENERGY_EQUIVALENT_IN_MEV
- HELION_MASS_IN_U
- HELION_MOLAR_MASS
- HELION_PROTON_MASS_RATIO
- HELION_RELATIVE_ATOMIC_MASS
- HELION_SHIELDING_SHIFT
- HERTZ_ATOMIC_MASS_UNIT_RELATIONSHIP
- HERTZ_ELECTRON_VOLT_RELATIONSHIP
- HERTZ_HARTREE_RELATIONSHIP

- HERTZ_INVERSE_METER_RELATIONSHIP
- HERTZ_JOULE_RELATIONSHIP
- HERTZ_KELVIN_RELATIONSHIP
- HERTZ_KILOGRAM_RELATIONSHIP
- HYPERFINE_TRANSITION_FREQUENCY_OF_CS_133
- INVERSE_FINE_STRUCTURE_CONSTANT
- INVERSE_METER_ATOMIC_MASS_UNIT_RELATIONSHIP
- INVERSE_METER_ELECTRON_VOLT_RELATIONSHIP
- INVERSE_METER_HARTREE_RELATIONSHIP
- INVERSE_METER_HERTZ_RELATIONSHIP
- INVERSE_METER_JOULE_RELATIONSHIP
- INVERSE_METER_KELVIN_RELATIONSHIP
- INVERSE_METER_KILOGRAM_RELATIONSHIP
- INVERSE_OF_CONDUCTANCE_QUANTUM
- JOSEPHSON_CONSTANT
- JOULE_ATOMIC_MASS_UNIT_RELATIONSHIP
- JOULE_ELECTRON_VOLT_RELATIONSHIP
- JOULE_HARTREE_RELATIONSHIP
- JOULE_HERTZ_RELATIONSHIP
- JOULE_INVERSE_METER_RELATIONSHIP
- JOULE_KELVIN_RELATIONSHIP
- JOULE_KILOGRAM_RELATIONSHIP
- KELVIN_ATOMIC_MASS_UNIT_RELATIONSHIP
- KELVIN_ELECTRON_VOLT_RELATIONSHIP
- KELVIN_HARTREE_RELATIONSHIP
- KELVIN_HERTZ_RELATIONSHIP
- KELVIN_INVERSE_METER_RELATIONSHIP
- KELVIN_JOULE_RELATIONSHIP
- KELVIN_KILOGRAM_RELATIONSHIP
- KILOGRAM_ATOMIC_MASS_UNIT_RELATIONSHIP
- KILOGRAM_ELECTRON_VOLT_RELATIONSHIP
- KILOGRAM_HARTREE_RELATIONSHIP
- KILOGRAM_HERTZ_RELATIONSHIP
- KILOGRAM_INVERSE_METER_RELATIONSHIP
- KILOGRAM_JOULE_RELATIONSHIP
- KILOGRAM_KELVIN_RELATIONSHIP
- LATTICE_PARAMETER_OF_SILICON
- LATTICE_SPACING_OF_IDEAL_SI_220

- LOSCHMIDT_CONSTANT_273_15_K_100_KPA
- LOSCHMIDT_CONSTANT_273_15_K_101_325_KPA
- LUMINOUS_EFFICACY
- MAG_FLUX_QUANTUM
- MOLAR_GAS_CONSTANT
- MOLAR_MASS_CONSTANT
- MOLAR_MASS_OF_CARBON_12
- MOLAR_PLANCK_CONSTANT
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_100_KPA
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_101_325_KPA
- MOLAR_VOLUME_OF_SILICON
- MOLYBDENUM_X_UNIT
- MUON_COMPTON_WAVELENGTH
- MUON_ELECTRON_MASS_RATIO
- MUON_G_FACTOR
- MUON_MAG_MOM
- MUON_MAG_MOM_ANOMALY
- MUON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- MUON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- MUON_MASS
- MUON_MASS_ENERGY_EQUIVALENT
- MUON_MASS_ENERGY_EQUIVALENT_IN_MEV
- MUON_MASS_IN_U
- MUON_MOLAR_MASS
- MUON_NEUTRON_MASS_RATIO
- MUON_PROTON_MAG_MOM_RATIO
- MUON_PROTON_MASS_RATIO
- MUON_TAU_MASS_RATIO
- NATURAL_UNIT_OF_ACTION
- NATURAL_UNIT_OF_ACTION_IN_EV_S
- NATURAL_UNIT_OF_ENERGY
- NATURAL_UNIT_OF_ENERGY_IN_MEV
- NATURAL_UNIT_OF_LENGTH
- NATURAL_UNIT_OF_MASS
- NATURAL_UNIT_OF_MOMENTUM
- NATURAL_UNIT_OF_MOMENTUM_IN_MEV_C
- NATURAL_UNIT_OF_TIME
- NATURAL_UNIT_OF_VELOCITY

- NEUTRON_COMPTON_WAVELENGTH
- NEUTRON_ELECTRON_MAG_MOM_RATIO
- NEUTRON_ELECTRON_MASS_RATIO
- NEUTRON_G_FACTOR
- NEUTRON_GYROMAG_RATIO
- NEUTRON_GYROMAG_RATIO_IN_MHZ_T
- NEUTRON_MAG_MOM
- NEUTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- NEUTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- NEUTRON_MASS
- NEUTRON_MASS_ENERGY_EQUIVALENT
- NEUTRON_MASS_ENERGY_EQUIVALENT_IN_MEV
- NEUTRON_MASS_IN_U
- NEUTRON_MOLAR_MASS
- NEUTRON_MUON_MASS_RATIO
- NEUTRON_PROTON_MAG_MOM_RATIO
- NEUTRON_PROTON_MASS_DIFFERENCE
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_IN_MEV
- NEUTRON_PROTON_MASS_DIFFERENCE_IN_U
- NEUTRON_PROTON_MASS_RATIO
- NEUTRON_RELATIVE_ATOMIC_MASS
- NEUTRON_TAU_MASS_RATIO
- NEUTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO
- NEWTONIAN_CONSTANT_OF_GRAVITATION
- NEWTONIAN_CONSTANT_OF_GRAVITATION_OVER_H_BAR_C
- NUCLEAR_MAGNETON
- NUCLEAR_MAGNETON_IN_EV_T
- NUCLEAR_MAGNETON_IN_INVERSE_METER_PER_TESLA
- NUCLEAR_MAGNETON_IN_K_T
- NUCLEAR_MAGNETON_IN_MHZ_T
- PLANCK_CONSTANT
- PLANCK_CONSTANT_IN_EV_HZ
- PLANCK_LENGTH
- PLANCK_MASS
- PLANCK_MASS_ENERGY_EQUIVALENT_IN_GEV
- PLANCK_TEMPERATURE
- PLANCK_TIME

- PROTON_CHARGE_TO_MASS_QUOTIENT
- PROTON_COMPTON_WAVELENGTH
- PROTON_ELECTRON_MASS_RATIO
- PROTON_G_FACTOR
- PROTON_GYROMAG_RATIO
- PROTON_GYROMAG_RATIO_IN_MHZ_T
- PROTON_MAG_MOM
- PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- PROTON_MAG_SHIELDING_CORRECTION
- PROTON_MASS
- PROTON_MASS_ENERGY_EQUIVALENT
- PROTON_MASS_ENERGY_EQUIVALENT_IN_MEV
- PROTON_MASS_IN_U
- PROTON_MOLAR_MASS
- PROTON_MUON_MASS_RATIO
- PROTON_NEUTRON_MAG_MOM_RATIO
- PROTON_NEUTRON_MASS_RATIO
- PROTON_RELATIVE_ATOMIC_MASS
- PROTON_RMS_CHARGE_RADIUS
- PROTON_TAU_MASS_RATIO
- QUANTUM_OF_CIRCULATION
- QUANTUM_OF_CIRCULATION_TIMES_2
- REDUCED_COMPTON_WAVELENGTH
- REDUCED_MUON_COMPTON_WAVELENGTH
- REDUCED_NEUTRON_COMPTON_WAVELENGTH
- REDUCED_PLANCK_CONSTANT
- REDUCED_PLANCK_CONSTANT_IN_EV_S
- REDUCED_PLANCK_CONSTANT_TIMES_C_IN_MEV_FM
- REDUCED_PROTON_COMPTON_WAVELENGTH
- REDUCED_TAU_COMPTON_WAVELENGTH
- RYDBERG_CONSTANT
- RYDBERG_CONSTANT_TIMES_C_IN_HZ
- RYDBERG_CONSTANT_TIMES_HC_IN_EV
- RYDBERG_CONSTANT_TIMES_HC_IN_J
- SACKUR_TETRODE_CONSTANT_1_K_100_KPA
- SACKUR_TETRODE_CONSTANT_1_K_101_325_KPA
- SECOND_RADIATION_CONSTANT

- SHIELDED_HELION_GYROMAG_RATIO
- SHIELDED_HELION_GYROMAG_RATIO_IN_MHZ_T
- SHIELDED_HELION_MAG_MOM
- SHIELDED_HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- SHIELDED_HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- SHIELDED_HELION_TO_PROTON_MAG_MOM_RATIO
- SHIELDED_HELION_TO_SHIELDED_PROTON_MAG_MOM_RATIO
- SHIELDED_PROTON_GYROMAG_RATIO
- SHIELDED_PROTON_GYROMAG_RATIO_IN_MHZ_T
- SHIELDED_PROTON_MAG_MOM
- SHIELDED_PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- SHIELDED_PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- SHIELDING_DIFFERENCE_OF_D_AND_P_IN_HD
- SHIELDING_DIFFERENCE_OF_T_AND_P_IN_HT
- SPEED_OF_LIGHT_IN_VACUUM
- STANDARD_ACCELERATION_OF_GRAVITY
- STANDARD_ATMOSPHERE
- STANDARD_STATE_PRESSURE
- STEFAN_BOLTZMANN_CONSTANT
- TAU_COMPTON_WAVELENGTH
- TAU_ELECTRON_MASS_RATIO
- TAU_ENERGY_EQUIVALENT
- TAU_MASS
- TAU_MASS_ENERGY_EQUIVALENT
- TAU_MASS_IN_U
- TAU_MOLAR_MASS
- TAU_MUON_MASS_RATIO
- TAU_NEUTRON_MASS_RATIO
- TAU_PROTON_MASS_RATIO
- THOMSON_CROSS_SECTION
- TRITON_ELECTRON_MASS_RATIO
- TRITON_G_FACTOR
- TRITON_MAG_MOM
- TRITON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- TRITON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- TRITON_MASS
- TRITON_MASS_ENERGY_EQUIVALENT
- TRITON_MASS_ENERGY_EQUIVALENT_IN_MEV

- TRITON_MASS_IN_U
- TRITON_MOLAR_MASS
- TRITON_PROTON_MASS_RATIO
- TRITON_RELATIVE_ATOMIC_MASS
- TRITON_TO_PROTON_MAG_MOM_RATIO
- UNIFIED_ATOMIC_MASS_UNIT
- VACUUM_ELECTRIC_PERMITTIVITY
- VACUUM_MAG_PERMEABILITY
- VON_KLITZING_CONSTANT
- WEAK_MIXING_ANGLE
- WIEN_FREQUENCY_DISPLACEMENT_LAW_CONSTANT
- WIEN_WAVELENGTH_DISPLACEMENT_LAW_CONSTANT
- W_TO_Z_MASS_RATIO

CODATA 2018

List of available constants:

- ALPHA_PARTICLE_ELECTRON_MASS_RATIO_2018
- ALPHA_PARTICLE_MASS_2018
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_2018
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- ALPHA_PARTICLE_MASS_IN_U_2018
- ALPHA_PARTICLE_MOLAR_MASS_2018
- ALPHA_PARTICLE_PROTON_MASS_RATIO_2018
- ALPHA_PARTICLE_RELATIVE_ATOMIC_MASS_2018
- ANGSTROM_STAR_2018
- ATOMIC_MASS_CONSTANT_2018
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_2018
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_IN_MEV_2018
- ATOMIC_MASS_UNIT_ELECTRON_VOLT_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_HARTREE_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_HERTZ_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_INVERSE_METER_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_JOULE_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_KELVIN_RELATIONSHIP_2018
- ATOMIC_MASS_UNIT_KILOGRAM_RELATIONSHIP_2018
- ATOMIC_UNIT_OF_1ST_HYPERPOLARIZABILITY_2018
- ATOMIC_UNIT_OF_2ND_HYPERPOLARIZABILITY_2018
- ATOMIC_UNIT_OF_ACTION_2018
- ATOMIC_UNIT_OF_CHARGE_2018

- ATOMIC_UNIT_OF_CHARGE_DENSITY_2018
- ATOMIC_UNIT_OF_CURRENT_2018
- ATOMIC_UNIT_OF_ELECTRIC_DIPOLE_MOM_2018
- ATOMIC_UNIT_OF_ELECTRIC_FIELD_2018
- ATOMIC_UNIT_OF_ELECTRIC_FIELD_GRADIENT_2018
- ATOMIC_UNIT_OF_ELECTRIC_POLARIZABILITY_2018
- ATOMIC_UNIT_OF_ELECTRIC_POTENTIAL_2018
- ATOMIC_UNIT_OF_ELECTRIC_QUADRUPOLE_MOM_2018
- ATOMIC_UNIT_OF_ENERGY_2018
- ATOMIC_UNIT_OF_FORCE_2018
- ATOMIC_UNIT_OF_LENGTH_2018
- ATOMIC_UNIT_OF_MAG_DIPOLE_MOM_2018
- ATOMIC_UNIT_OF_MAG_FLUX_DENSITY_2018
- ATOMIC_UNIT_OF_MAGNETIZABILITY_2018
- ATOMIC_UNIT_OF_MASS_2018
- ATOMIC_UNIT_OF_MOMENTUM_2018
- ATOMIC_UNIT_OF_PERMITTIVITY_2018
- ATOMIC_UNIT_OF_TIME_2018
- ATOMIC_UNIT_OF_VELOCITY_2018
- AVOGADRO_CONSTANT_2018
- BOHR_MAGNETON_2018
- BOHR_MAGNETON_IN_EV_T_2018
- BOHR_MAGNETON_IN_HZ_T_2018
- BOHR_MAGNETON_IN_INVERSE_METER_PER_TESLA_2018
- BOHR_MAGNETON_IN_K_T_2018
- BOHR_RADIUS_2018
- BOLTZMANN_CONSTANT_2018
- BOLTZMANN_CONSTANT_IN_EV_K_2018
- BOLTZMANN_CONSTANT_IN_HZ_K_2018
- BOLTZMANN_CONSTANT_IN_INVERSE_METER_PER_KELVIN_2018
- CHARACTERISTIC_IMPEDANCE_OF_VACUUM_2018
- CLASSICAL_ELECTRON_RADIUS_2018
- COMPTON_WAVELENGTH_2018
- CONDUCTANCE_QUANTUM_2018
- CONVENTIONAL_VALUE_OF_AMPERE_90_2018
- CONVENTIONAL_VALUE_OF_COULOMB_90_2018
- CONVENTIONAL_VALUE_OF_FARAD_90_2018
- CONVENTIONAL_VALUE_OF_HENRY_90_2018

- CONVENTIONAL_VALUE_OF_JOSEPHSON_CONSTANT_2018
- CONVENTIONAL_VALUE_OF_OHM_90_2018
- CONVENTIONAL_VALUE_OF_VOLT_90_2018
- CONVENTIONAL_VALUE_OF_VON_KLITZING_CONSTANT_2018
- CONVENTIONAL_VALUE_OF_WATT_90_2018
- COPPER_X_UNIT_2018
- DEUTERON_ELECTRON_MAG_MOM_RATIO_2018
- DEUTERON_ELECTRON_MASS_RATIO_2018
- DEUTERON_G_FACTOR_2018
- DEUTERON_MAG_MOM_2018
- DEUTERON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- DEUTERON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- DEUTERON_MASS_2018
- DEUTERON_MASS_ENERGY_EQUIVALENT_2018
- DEUTERON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- DEUTERON_MASS_IN_U_2018
- DEUTERON_MOLAR_MASS_2018
- DEUTERON_NEUTRON_MAG_MOM_RATIO_2018
- DEUTERON_PROTON_MAG_MOM_RATIO_2018
- DEUTERON_PROTON_MASS_RATIO_2018
- DEUTERON_RELATIVE_ATOMIC_MASS_2018
- DEUTERON_RMS_CHARGE_RADIUS_2018
- ELECTRON_CHARGE_TO_MASS_QUOTIENT_2018
- ELECTRON_DEUTERON_MAG_MOM_RATIO_2018
- ELECTRON_DEUTERON_MASS_RATIO_2018
- ELECTRON_G_FACTOR_2018
- ELECTRON_GYROMAG_RATIO_2018
- ELECTRON_GYROMAG_RATIO_IN_MHZ_T_2018
- ELECTRON_HELION_MASS_RATIO_2018
- ELECTRON_MAG_MOM_2018
- ELECTRON_MAG_MOM_ANOMALY_2018
- ELECTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- ELECTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- ELECTRON_MASS_2018
- ELECTRON_MASS_ENERGY_EQUIVALENT_2018
- ELECTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- ELECTRON_MASS_IN_U_2018
- ELECTRON_MOLAR_MASS_2018

- ELECTRON_MUON_MAG_MOM_RATIO_2018
- ELECTRON_MUON_MASS_RATIO_2018
- ELECTRON_NEUTRON_MAG_MOM_RATIO_2018
- ELECTRON_NEUTRON_MASS_RATIO_2018
- ELECTRON_PROTON_MAG_MOM_RATIO_2018
- ELECTRON_PROTON_MASS_RATIO_2018
- ELECTRON_RELATIVE_ATOMIC_MASS_2018
- ELECTRON_TAU_MASS_RATIO_2018
- ELECTRON_TO_ALPHA_PARTICLE_MASS_RATIO_2018
- ELECTRON_TO_SHIELDED_HELION_MAG_MOM_RATIO_2018
- ELECTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2018
- ELECTRON_TRITON_MASS_RATIO_2018
- ELECTRON_VOLT_2018
- ELECTRON_VOLT_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- ELECTRON_VOLT_HARTREE_RELATIONSHIP_2018
- ELECTRON_VOLT_HERTZ_RELATIONSHIP_2018
- ELECTRON_VOLT_INVERSE_METER_RELATIONSHIP_2018
- ELECTRON_VOLT_JOULE_RELATIONSHIP_2018
- ELECTRON_VOLT_KELVIN_RELATIONSHIP_2018
- ELECTRON_VOLT_KILOGRAM_RELATIONSHIP_2018
- ELEMENTARY_CHARGE_2018
- ELEMENTARY_CHARGE_OVER_H_BAR_2018
- FARADAY_CONSTANT_2018
- FERMI_COUPLING_CONSTANT_2018
- FINE_STRUCTURE_CONSTANT_2018
- FIRST_RADIATION_CONSTANT_2018
- FIRST_RADIATION_CONSTANT_FOR_SPECTRAL_RADIANCE_2018
- HARTREE_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- HARTREE_ELECTRON_VOLT_RELATIONSHIP_2018
- HARTREE_ENERGY_2018
- HARTREE_ENERGY_IN_EV_2018
- HARTREE_HERTZ_RELATIONSHIP_2018
- HARTREE_INVERSE_METER_RELATIONSHIP_2018
- HARTREE_JOULE_RELATIONSHIP_2018
- HARTREE_KELVIN_RELATIONSHIP_2018
- HARTREE_KILOGRAM_RELATIONSHIP_2018
- HELION_ELECTRON_MASS_RATIO_2018
- HELION_G_FACTOR_2018

- HELION_MAG_MOM_2018
- HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- HELION_MASS_2018
- HELION_MASS_ENERGY_EQUIVALENT_2018
- HELION_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- HELION_MASS_IN_U_2018
- HELION_MOLAR_MASS_2018
- HELION_PROTON_MASS_RATIO_2018
- HELION_RELATIVE_ATOMIC_MASS_2018
- HELION_SHIELDING_SHIFT_2018
- HERTZ_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- HERTZ_ELECTRON_VOLT_RELATIONSHIP_2018
- HERTZ_HARTREE_RELATIONSHIP_2018
- HERTZ_INVERSE_METER_RELATIONSHIP_2018
- HERTZ_JOULE_RELATIONSHIP_2018
- HERTZ_KELVIN_RELATIONSHIP_2018
- HERTZ_KILOGRAM_RELATIONSHIP_2018
- HYPERFINE_TRANSITION_FREQUENCY_OF_CS_133_2018
- INVERSE_FINE_STRUCTURE_CONSTANT_2018
- INVERSE_METER_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- INVERSE_METER_ELECTRON_VOLT_RELATIONSHIP_2018
- INVERSE_METER_HARTREE_RELATIONSHIP_2018
- INVERSE_METER_HERTZ_RELATIONSHIP_2018
- INVERSE_METER_JOULE_RELATIONSHIP_2018
- INVERSE_METER_KELVIN_RELATIONSHIP_2018
- INVERSE_METER_KILOGRAM_RELATIONSHIP_2018
- INVERSE_OF_CONDUCTANCE_QUANTUM_2018
- JOSEPHSON_CONSTANT_2018
- JOULE_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- JOULE_ELECTRON_VOLT_RELATIONSHIP_2018
- JOULE_HARTREE_RELATIONSHIP_2018
- JOULE_HERTZ_RELATIONSHIP_2018
- JOULE_INVERSE_METER_RELATIONSHIP_2018
- JOULE_KELVIN_RELATIONSHIP_2018
- JOULE_KILOGRAM_RELATIONSHIP_2018
- KELVIN_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- KELVIN_ELECTRON_VOLT_RELATIONSHIP_2018

- KELVIN_HARTREE_RELATIONSHIP_2018
- KELVIN_HERTZ_RELATIONSHIP_2018
- KELVIN_INVERSE_METER_RELATIONSHIP_2018
- KELVIN_JOULE_RELATIONSHIP_2018
- KELVIN_KILOGRAM_RELATIONSHIP_2018
- KILOGRAM_ATOMIC_MASS_UNIT_RELATIONSHIP_2018
- KILOGRAM_ELECTRON_VOLT_RELATIONSHIP_2018
- KILOGRAM_HARTREE_RELATIONSHIP_2018
- KILOGRAM_HERTZ_RELATIONSHIP_2018
- KILOGRAM_INVERSE_METER_RELATIONSHIP_2018
- KILOGRAM_JOULE_RELATIONSHIP_2018
- KILOGRAM_KELVIN_RELATIONSHIP_2018
- LATTICE_PARAMETER_OF_SILICON_2018
- LATTICE_SPACING_OF_IDEAL_SI_220_2018
- LOSCHMIDT_CONSTANT_273_15_K_100_KPA_2018
- LOSCHMIDT_CONSTANT_273_15_K_101_325_KPA_2018
- LUMINOUS_EFFICACY_2018
- MAG_FLUX_QUANTUM_2018
- MOLAR_GAS_CONSTANT_2018
- MOLAR_MASS_CONSTANT_2018
- MOLAR_MASS_OF_CARBON_12_2018
- MOLAR_PLANCK_CONSTANT_2018
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_100_KPA_2018
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_101_325_KPA_2018
- MOLAR_VOLUME_OF_SILICON_2018
- MOLYBDENUM_X_UNIT_2018
- MUON_COMPTON_WAVELENGTH_2018
- MUON_ELECTRON_MASS_RATIO_2018
- MUON_G_FACTOR_2018
- MUON_MAG_MOM_2018
- MUON_MAG_MOM_ANOMALY_2018
- MUON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- MUON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- MUON_MASS_2018
- MUON_MASS_ENERGY_EQUIVALENT_2018
- MUON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- MUON_MASS_IN_U_2018
- MUON_MOLAR_MASS_2018

- MUON_NEUTRON_MASS_RATIO_2018
- MUON_PROTON_MAG_MOM_RATIO_2018
- MUON_PROTON_MASS_RATIO_2018
- MUON_TAU_MASS_RATIO_2018
- NATURAL_UNIT_OF_ACTION_2018
- NATURAL_UNIT_OF_ACTION_IN_EV_S_2018
- NATURAL_UNIT_OF_ENERGY_2018
- NATURAL_UNIT_OF_ENERGY_IN_MEV_2018
- NATURAL_UNIT_OF_LENGTH_2018
- NATURAL_UNIT_OF_MASS_2018
- NATURAL_UNIT_OF_MOMENTUM_2018
- NATURAL_UNIT_OF_MOMENTUM_IN_MEV_C_2018
- NATURAL_UNIT_OF_TIME_2018
- NATURAL_UNIT_OF_VELOCITY_2018
- NEUTRON_COMPTON_WAVELENGTH_2018
- NEUTRON_ELECTRON_MAG_MOM_RATIO_2018
- NEUTRON_ELECTRON_MASS_RATIO_2018
- NEUTRON_G_FACTOR_2018
- NEUTRON_GYROMAG_RATIO_2018
- NEUTRON_GYROMAG_RATIO_IN_MHZ_T_2018
- NEUTRON_MAG_MOM_2018
- NEUTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- NEUTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- NEUTRON_MASS_2018
- NEUTRON_MASS_ENERGY_EQUIVALENT_2018
- NEUTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- NEUTRON_MASS_IN_U_2018
- NEUTRON_MOLAR_MASS_2018
- NEUTRON_MUON_MASS_RATIO_2018
- NEUTRON_PROTON_MAG_MOM_RATIO_2018
- NEUTRON_PROTON_MASS_DIFFERENCE_2018
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_2018
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_IN_MEV_2018
- NEUTRON_PROTON_MASS_DIFFERENCE_IN_U_2018
- NEUTRON_PROTON_MASS_RATIO_2018
- NEUTRON_RELATIVE_ATOMIC_MASS_2018
- NEUTRON_TAU_MASS_RATIO_2018
- NEUTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2018

- NEWTONIAN_CONSTANT_OF_GRAVITATION_2018
- NEWTONIAN_CONSTANT_OF_GRAVITATION_OVER_H_BAR_C_2018
- NUCLEAR_MAGNETON_2018
- NUCLEAR_MAGNETON_IN_EV_T_2018
- NUCLEAR_MAGNETON_IN_INVERSE_METER_PER_TESLA_2018
- NUCLEAR_MAGNETON_IN_K_T_2018
- NUCLEAR_MAGNETON_IN_MHZ_T_2018
- PLANCK_CONSTANT_2018
- PLANCK_CONSTANT_IN_EV_HZ_2018
- PLANCK_LENGTH_2018
- PLANCK_MASS_2018
- PLANCK_MASS_ENERGY_EQUIVALENT_IN_GEV_2018
- PLANCK_TEMPERATURE_2018
- PLANCK_TIME_2018
- PROTON_CHARGE_TO_MASS_QUOTIENT_2018
- PROTON_COMPTON_WAVELENGTH_2018
- PROTON_ELECTRON_MASS_RATIO_2018
- PROTON_G_FACTOR_2018
- PROTON_GYROMAG_RATIO_2018
- PROTON_GYROMAG_RATIO_IN_MHZ_T_2018
- PROTON_MAG_MOM_2018
- PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- PROTON_MAG_SHIELDING_CORRECTION_2018
- PROTON_MASS_2018
- PROTON_MASS_ENERGY_EQUIVALENT_2018
- PROTON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- PROTON_MASS_IN_U_2018
- PROTON_MOLAR_MASS_2018
- PROTON_MUON_MASS_RATIO_2018
- PROTON_NEUTRON_MAG_MOM_RATIO_2018
- PROTON_NEUTRON_MASS_RATIO_2018
- PROTON_RELATIVE_ATOMIC_MASS_2018
- PROTON_RMS_CHARGE_RADIUS_2018
- PROTON_TAU_MASS_RATIO_2018
- QUANTUM_OF_CIRCULATION_2018
- QUANTUM_OF_CIRCULATION_TIMES_2_2018
- REDUCED_COMPTON_WAVELENGTH_2018

- REDUCED_MUON_COMPTON_WAVELENGTH_2018
- REDUCED_NEUTRON_COMPTON_WAVELENGTH_2018
- REDUCED_PLANCK_CONSTANT_2018
- REDUCED_PLANCK_CONSTANT_IN_EV_S_2018
- REDUCED_PLANCK_CONSTANT_TIMES_C_IN_MEV_FM_2018
- REDUCED_PROTON_COMPTON_WAVELENGTH_2018
- REDUCED_TAU_COMPTON_WAVELENGTH_2018
- RYDBERG_CONSTANT_2018
- RYDBERG_CONSTANT_TIMES_C_IN_HZ_2018
- RYDBERG_CONSTANT_TIMES_HC_IN_EV_2018
- RYDBERG_CONSTANT_TIMES_HC_IN_J_2018
- SACKUR_TETRODE_CONSTANT_1_K_100_KPA_2018
- SACKUR_TETRODE_CONSTANT_1_K_101_325_KPA_2018
- SECOND_RADIATION_CONSTANT_2018
- SHIELDED_HELION_GYROMAG_RATIO_2018
- SHIELDED_HELION_GYROMAG_RATIO_IN_MHZ_T_2018
- SHIELDED_HELION_MAG_MOM_2018
- SHIELDED_HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- SHIELDED_HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- SHIELDED_HELION_TO_PROTON_MAG_MOM_RATIO_2018
- SHIELDED_HELION_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2018
- SHIELDED_PROTON_GYROMAG_RATIO_2018
- SHIELDED_PROTON_GYROMAG_RATIO_IN_MHZ_T_2018
- SHIELDED_PROTON_MAG_MOM_2018
- SHIELDED_PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- SHIELDED_PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- SHIELDING_DIFFERENCE_OF_D_AND_P_IN_HD_2018
- SHIELDING_DIFFERENCE_OF_T_AND_P_IN_HT_2018
- SPEED_OF_LIGHT_IN_VACUUM_2018
- STANDARD_ACCELERATION_OF_GRAVITY_2018
- STANDARD_ATMOSPHERE_2018
- STANDARD_STATE_PRESSURE_2018
- STEFAN_BOLTZMANN_CONSTANT_2018
- TAU_COMPTON_WAVELENGTH_2018
- TAU_ELECTRON_MASS_RATIO_2018
- TAU_ENERGY_EQUIVALENT_2018
- TAU_MASS_2018
- TAU_MASS_ENERGY_EQUIVALENT_2018

- TAU_MASS_IN_U_2018
- TAU_MOLAR_MASS_2018
- TAU_MUON_MASS_RATIO_2018
- TAU_NEUTRON_MASS_RATIO_2018
- TAU_PROTON_MASS_RATIO_2018
- THOMSON_CROSS_SECTION_2018
- TRITON_ELECTRON_MASS_RATIO_2018
- TRITON_G_FACTOR_2018
- TRITON_MAG_MOM_2018
- TRITON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2018
- TRITON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2018
- TRITON_MASS_2018
- TRITON_MASS_ENERGY_EQUIVALENT_2018
- TRITON_MASS_ENERGY_EQUIVALENT_IN_MEV_2018
- TRITON_MASS_IN_U_2018
- TRITON_MOLAR_MASS_2018
- TRITON_PROTON_MASS_RATIO_2018
- TRITON_RELATIVE_ATOMIC_MASS_2018
- TRITON_TO_PROTON_MAG_MOM_RATIO_2018
- UNIFIED_ATOMIC_MASS_UNIT_2018
- VACUUM_ELECTRIC_PERMITTIVITY_2018
- VACUUM_MAG_PERMEABILITY_2018
- VON_KLITZING_CONSTANT_2018
- WEAK_MIXING_ANGLE_2018
- WIEN_FREQUENCY_DISPLACEMENT_LAW_CONSTANT_2018
- WIEN_WAVELENGTH_DISPLACEMENT_LAW_CONSTANT_2018
- W_TO_Z_MASS_RATIO_2018

CODATA 2014

List of available constants:

- LATTICE_SPACING_OF_SILICON_2014
- ALPHA_PARTICLE_ELECTRON_MASS_RATIO_2014
- ALPHA_PARTICLE_MASS_2014
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_2014
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- ALPHA_PARTICLE_MASS_IN_U_2014
- ALPHA_PARTICLE_MOLAR_MASS_2014
- ALPHA_PARTICLE_PROTON_MASS_RATIO_2014
- ANGSTROM_STAR_2014

- ATOMIC_MASS_CONSTANT_2014
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_2014
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_IN_MEV_2014
- ATOMIC_MASS_UNIT_ELECTRON_VOLT_RELATIONSHIP_2014
- ATOMIC_MASS_UNIT_HARTREE_RELATIONSHIP_2014
- ATOMIC_MASS_UNIT_HERTZ_RELATIONSHIP_2014
- ATOMIC_MASS_UNIT_INVERSE_METER_RELATIONSHIP_2014
- ATOMIC_MASS_UNIT_JOULE_RELATIONSHIP_2014
- ATOMIC_MASS_UNIT_KELVIN_RELATIONSHIP_2014
- ATOMIC_MASS_UNIT_KILOGRAM_RELATIONSHIP_2014
- ATOMIC_UNIT_OF_1ST_HYPERPOLARIZABILITY_2014
- ATOMIC_UNIT_OF_2ND_HYPERPOLARIZABILITY_2014
- ATOMIC_UNIT_OF_ACTION_2014
- ATOMIC_UNIT_OF_CHARGE_2014
- ATOMIC_UNIT_OF_CHARGE_DENSITY_2014
- ATOMIC_UNIT_OF_CURRENT_2014
- ATOMIC_UNIT_OF_ELECTRIC_DIPOLE_MOM_2014
- ATOMIC_UNIT_OF_ELECTRIC_FIELD_2014
- ATOMIC_UNIT_OF_ELECTRIC_FIELD_GRADIENT_2014
- ATOMIC_UNIT_OF_ELECTRIC_POLARIZABILITY_2014
- ATOMIC_UNIT_OF_ELECTRIC_POTENTIAL_2014
- ATOMIC_UNIT_OF_ELECTRIC_QUADRUPOLE_MOM_2014
- ATOMIC_UNIT_OF_ENERGY_2014
- ATOMIC_UNIT_OF_FORCE_2014
- ATOMIC_UNIT_OF_LENGTH_2014
- ATOMIC_UNIT_OF_MAG_DIPOLE_MOM_2014
- ATOMIC_UNIT_OF_MAG_FLUX_DENSITY_2014
- ATOMIC_UNIT_OF_MAGNETIZABILITY_2014
- ATOMIC_UNIT_OF_MASS_2014
- ATOMIC_UNIT_OF_MOMUM_2014
- ATOMIC_UNIT_OF_PERMITTIVITY_2014
- ATOMIC_UNIT_OF_TIME_2014
- ATOMIC_UNIT_OF_VELOCITY_2014
- AVOGADRO_CONSTANT_2014
- BOHR_MAGNETON_2014
- BOHR_MAGNETON_IN_EV_T_2014
- BOHR_MAGNETON_IN_HZ_T_2014
- BOHR_MAGNETON_IN_INVERSE_METERS_PER_TESLA_2014

- BOHR_MAGNETON_IN_K_T_2014
- BOHR_RADIUS_2014
- BOLTZMANN_CONSTANT_2014
- BOLTZMANN_CONSTANT_IN_EV_K_2014
- BOLTZMANN_CONSTANT_IN_HZ_K_2014
- BOLTZMANN_CONSTANT_IN_INVERSE_METERS_PER_KELVIN_2014
- CHARACTERISTIC_IMPEDANCE_OF_VACUUM_2014
- CLASSICAL_ELECTRON_RADIUS_2014
- COMPTON_WAVELENGTH_2014
- COMPTON_WAVELENGTH_OVER_2_PI_2014
- CONDUCTANCE_QUANTUM_2014
- CONVENTIONAL_VALUE_OF_JOSEPHSON_CONSTANT_2014
- CONVENTIONAL_VALUE_OF_VON_KLITZING_CONSTANT_2014
- CU_X_UNIT_2014
- DEUTERON_ELECTRON_MAG_MOM_RATIO_2014
- DEUTERON_ELECTRON_MASS_RATIO_2014
- DEUTERON_G_FACTOR_2014
- DEUTERON_MAG_MOM_2014
- DEUTERON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- DEUTERON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- DEUTERON_MASS_2014
- DEUTERON_MASS_ENERGY_EQUIVALENT_2014
- DEUTERON_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- DEUTERON_MASS_IN_U_2014
- DEUTERON_MOLAR_MASS_2014
- DEUTERON_NEUTRON_MAG_MOM_RATIO_2014
- DEUTERON_PROTON_MAG_MOM_RATIO_2014
- DEUTERON_PROTON_MASS_RATIO_2014
- DEUTERON_RMS_CHARGE_RADIUS_2014
- ELECTRIC_CONSTANT_2014
- ELECTRON_CHARGE_TO_MASS_QUOTIENT_2014
- ELECTRON_DEUTERON_MAG_MOM_RATIO_2014
- ELECTRON_DEUTERON_MASS_RATIO_2014
- ELECTRON_G_FACTOR_2014
- ELECTRON_GYROMAG_RATIO_2014
- ELECTRON_GYROMAG_RATIO_OVER_2_PI_2014
- ELECTRON_HELION_MASS_RATIO_2014
- ELECTRON_MAG_MOM_2014

- ELECTRON_MAG_MOM_ANOMALY_2014
- ELECTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- ELECTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- ELECTRON_MASS_2014
- ELECTRON_MASS_ENERGY_EQUIVALENT_2014
- ELECTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- ELECTRON_MASS_IN_U_2014
- ELECTRON_MOLAR_MASS_2014
- ELECTRON_MUON_MAG_MOM_RATIO_2014
- ELECTRON_MUON_MASS_RATIO_2014
- ELECTRON_NEUTRON_MAG_MOM_RATIO_2014
- ELECTRON_NEUTRON_MASS_RATIO_2014
- ELECTRON_PROTON_MAG_MOM_RATIO_2014
- ELECTRON_PROTON_MASS_RATIO_2014
- ELECTRON_TAU_MASS_RATIO_2014
- ELECTRON_TO_ALPHA_PARTICLE_MASS_RATIO_2014
- ELECTRON_TO_SHIELDED_HELION_MAG_MOM_RATIO_2014
- ELECTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2014
- ELECTRON_TRITON_MASS_RATIO_2014
- ELECTRON_VOLT_2014
- ELECTRON_VOLT_ATOMIC_MASS_UNIT_RELATIONSHIP_2014
- ELECTRON_VOLT_HARTREE_RELATIONSHIP_2014
- ELECTRON_VOLT_HERTZ_RELATIONSHIP_2014
- ELECTRON_VOLT_INVERSE_METER_RELATIONSHIP_2014
- ELECTRON_VOLT_JOULE_RELATIONSHIP_2014
- ELECTRON_VOLT_KELVIN_RELATIONSHIP_2014
- ELECTRON_VOLT_KILOGRAM_RELATIONSHIP_2014
- ELEMENTARY_CHARGE_2014
- ELEMENTARY_CHARGE_OVER_H_2014
- FARADAY_CONSTANT_2014
- FARADAY_CONSTANT_FOR_CONVENTIONAL_ELECTRIC_CURRENT_2014
- FERMI_COUPLING_CONSTANT_2014
- FINE_STRUCTURE_CONSTANT_2014
- FIRST_RADIATION_CONSTANT_2014
- FIRST_RADIATION_CONSTANT_FOR_SPECTRAL_RADIANCE_2014
- HARTREE_ATOMIC_MASS_UNIT_RELATIONSHIP_2014
- HARTREE_ELECTRON_VOLT_RELATIONSHIP_2014
- HARTREE_ENERGY_2014

- HARTREE_ENERGY_IN_EV_2014
- HARTREE_HERTZ_RELATIONSHIP_2014
- HARTREE_INVERSE_METER_RELATIONSHIP_2014
- HARTREE_JOULE_RELATIONSHIP_2014
- HARTREE_KELVIN_RELATIONSHIP_2014
- HARTREE_KILOGRAM_RELATIONSHIP_2014
- HELION_ELECTRON_MASS_RATIO_2014
- HELION_G_FACTOR_2014
- HELION_MAG_MOM_2014
- HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- HELION_MASS_2014
- HELION_MASS_ENERGY_EQUIVALENT_2014
- HELION_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- HELION_MASS_IN_U_2014
- HELION_MOLAR_MASS_2014
- HELION_PROTON_MASS_RATIO_2014
- HERTZ_ATOMIC_MASS_UNIT_RELATIONSHIP_2014
- HERTZ_ELECTRON_VOLT_RELATIONSHIP_2014
- HERTZ_HARTREE_RELATIONSHIP_2014
- HERTZ_INVERSE_METER_RELATIONSHIP_2014
- HERTZ_JOULE_RELATIONSHIP_2014
- HERTZ_KELVIN_RELATIONSHIP_2014
- HERTZ_KILOGRAM_RELATIONSHIP_2014
- INVERSE_FINE_STRUCTURE_CONSTANT_2014
- INVERSE_METER_ATOMIC_MASS_UNIT_RELATIONSHIP_2014
- INVERSE_METER_ELECTRON_VOLT_RELATIONSHIP_2014
- INVERSE_METER_HARTREE_RELATIONSHIP_2014
- INVERSE_METER_HERTZ_RELATIONSHIP_2014
- INVERSE_METER_JOULE_RELATIONSHIP_2014
- INVERSE_METER_KELVIN_RELATIONSHIP_2014
- INVERSE_METER_KILOGRAM_RELATIONSHIP_2014
- INVERSE_OF_CONDUCTANCE_QUANTUM_2014
- JOSEPHSON_CONSTANT_2014
- JOULE_ATOMIC_MASS_UNIT_RELATIONSHIP_2014
- JOULE_ELECTRON_VOLT_RELATIONSHIP_2014
- JOULE_HARTREE_RELATIONSHIP_2014
- JOULE_HERTZ_RELATIONSHIP_2014

- JOULE_INVERSE_METER_RELATIONSHIP_2014
- JOULE_KELVIN_RELATIONSHIP_2014
- JOULE_KILOGRAM_RELATIONSHIP_2014
- KELVIN_ATOMIC_MASS_UNIT_RELATIONSHIP_2014
- KELVIN_ELECTRON_VOLT_RELATIONSHIP_2014
- KELVIN_HARTREE_RELATIONSHIP_2014
- KELVIN_HERTZ_RELATIONSHIP_2014
- KELVIN_INVERSE_METER_RELATIONSHIP_2014
- KELVIN_JOULE_RELATIONSHIP_2014
- KELVIN_KILOGRAM_RELATIONSHIP_2014
- KILOGRAM_ATOMIC_MASS_UNIT_RELATIONSHIP_2014
- KILOGRAM_ELECTRON_VOLT_RELATIONSHIP_2014
- KILOGRAM_HARTREE_RELATIONSHIP_2014
- KILOGRAM_HERTZ_RELATIONSHIP_2014
- KILOGRAM_INVERSE_METER_RELATIONSHIP_2014
- KILOGRAM_JOULE_RELATIONSHIP_2014
- KILOGRAM_KELVIN_RELATIONSHIP_2014
- LATTICE_PARAMETER_OF_SILICON_2014
- LOSCHMIDT_CONSTANT_273_15_K_100_KPA_2014
- LOSCHMIDT_CONSTANT_273_15_K_101_325_KPA_2014
- MAG_CONSTANT_2014
- MAG_FLUX_QUANTUM_2014
- MOLAR_GAS_CONSTANT_2014
- MOLAR_MASS_CONSTANT_2014
- MOLAR_MASS_OF_CARBON_12_2014
- MOLAR_PLANCK_CONSTANT_2014
- MOLAR_PLANCK_CONSTANT_TIMES_C_2014
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_100_KPA_2014
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_101_325_KPA_2014
- MOLAR_VOLUME_OF_SILICON_2014
- MO_X_UNIT_2014
- MUON_COMPTON_WAVELENGTH_2014
- MUON_COMPTON_WAVELENGTH_OVER_2_PI_2014
- MUON_ELECTRON_MASS_RATIO_2014
- MUON_G_FACTOR_2014
- MUON_MAG_MOM_2014
- MUON_MAG_MOM_ANOMALY_2014
- MUON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014

- MUON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- MUON_MASS_2014
- MUON_MASS_ENERGY_EQUIVALENT_2014
- MUON_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- MUON_MASS_IN_U_2014
- MUON_MOLAR_MASS_2014
- MUON_NEUTRON_MASS_RATIO_2014
- MUON_PROTON_MAG_MOM_RATIO_2014
- MUON_PROTON_MASS_RATIO_2014
- MUON_TAU_MASS_RATIO_2014
- NATURAL_UNIT_OF_ACTION_2014
- NATURAL_UNIT_OF_ACTION_IN_EV_S_2014
- NATURAL_UNIT_OF_ENERGY_2014
- NATURAL_UNIT_OF_ENERGY_IN_MEV_2014
- NATURAL_UNIT_OF_LENGTH_2014
- NATURAL_UNIT_OF_MASS_2014
- NATURAL_UNIT_OF_MOMUM_2014
- NATURAL_UNIT_OF_MOMUM_IN_MEV_C_2014
- NATURAL_UNIT_OF_TIME_2014
- NATURAL_UNIT_OF_VELOCITY_2014
- NEUTRON_COMPTON_WAVELENGTH_2014
- NEUTRON_COMPTON_WAVELENGTH_OVER_2_PI_2014
- NEUTRON_ELECTRON_MAG_MOM_RATIO_2014
- NEUTRON_ELECTRON_MASS_RATIO_2014
- NEUTRON_G_FACTOR_2014
- NEUTRON_GYROMAG_RATIO_2014
- NEUTRON_GYROMAG_RATIO_OVER_2_PI_2014
- NEUTRON_MAG_MOM_2014
- NEUTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- NEUTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- NEUTRON_MASS_2014
- NEUTRON_MASS_ENERGY_EQUIVALENT_2014
- NEUTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- NEUTRON_MASS_IN_U_2014
- NEUTRON_MOLAR_MASS_2014
- NEUTRON_MUON_MASS_RATIO_2014
- NEUTRON_PROTON_MAG_MOM_RATIO_2014
- NEUTRON_PROTON_MASS_DIFFERENCE_2014

- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_2014
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_IN_MEV_2014
- NEUTRON_PROTON_MASS_DIFFERENCE_IN_U_2014
- NEUTRON_PROTON_MASS_RATIO_2014
- NEUTRON_TAU_MASS_RATIO_2014
- NEUTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2014
- NEWTONIAN_CONSTANT_OF_GRAVITATION_2014
- NEWTONIAN_CONSTANT_OF_GRAVITATION_OVER_H_BAR_C_2014
- NUCLEAR_MAGNETON_2014
- NUCLEAR_MAGNETON_IN_EV_T_2014
- NUCLEAR_MAGNETON_IN_INVERSE_METERS_PER_TESLA_2014
- NUCLEAR_MAGNETON_IN_K_T_2014
- NUCLEAR_MAGNETON_IN_MHZ_T_2014
- PLANCK_CONSTANT_2014
- PLANCK_CONSTANT_IN_EV_S_2014
- PLANCK_CONSTANT_OVER_2_PI_2014
- PLANCK_CONSTANT_OVER_2_PI_IN_EV_S_2014
- PLANCK_CONSTANT_OVER_2_PI_TIMES_C_IN_MEV_FM_2014
- PLANCK_LENGTH_2014
- PLANCK_MASS_2014
- PLANCK_MASS_ENERGY_EQUIVALENT_IN_GEV_2014
- PLANCK_TEMPERATURE_2014
- PLANCK_TIME_2014
- PROTON_CHARGE_TO_MASS_QUOTIENT_2014
- PROTON_COMPTON_WAVELENGTH_2014
- PROTON_COMPTON_WAVELENGTH_OVER_2_PI_2014
- PROTON_ELECTRON_MASS_RATIO_2014
- PROTON_G_FACTOR_2014
- PROTON_GYROMAG_RATIO_2014
- PROTON_GYROMAG_RATIO_OVER_2_PI_2014
- PROTON_MAG_MOM_2014
- PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- PROTON_MAG_SHIELDING_CORRECTION_2014
- PROTON_MASS_2014
- PROTON_MASS_ENERGY_EQUIVALENT_2014
- PROTON_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- PROTON_MASS_IN_U_2014

- PROTON_MOLAR_MASS_2014
- PROTON_MUON_MASS_RATIO_2014
- PROTON_NEUTRON_MAG_MOM_RATIO_2014
- PROTON_NEUTRON_MASS_RATIO_2014
- PROTON_RMS_CHARGE_RADIUS_2014
- PROTON_TAU_MASS_RATIO_2014
- QUANTUM_OF_CIRCULATION_2014
- QUANTUM_OF_CIRCULATION_TIMES_2_2014
- RYDBERG_CONSTANT_2014
- RYDBERG_CONSTANT_TIMES_C_IN_HZ_2014
- RYDBERG_CONSTANT_TIMES_HC_IN_EV_2014
- RYDBERG_CONSTANT_TIMES_HC_IN_J_2014
- SACKUR_TETRODE_CONSTANT_1_K_100_KPA_2014
- SACKUR_TETRODE_CONSTANT_1_K_101_325_KPA_2014
- SECOND_RADIATION_CONSTANT_2014
- SHIELDED_HELION_GYROMAG_RATIO_2014
- SHIELDED_HELION_GYROMAG_RATIO_OVER_2_PI_2014
- SHIELDED_HELION_MAG_MOM_2014
- SHIELDED_HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- SHIELDED_HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- SHIELDED_HELION_TO_PROTON_MAG_MOM_RATIO_2014
- SHIELDED_HELION_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2014
- SHIELDED_PROTON_GYROMAG_RATIO_2014
- SHIELDED_PROTON_GYROMAG_RATIO_OVER_2_PI_2014
- SHIELDED_PROTON_MAG_MOM_2014
- SHIELDED_PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- SHIELDED_PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- SPEED_OF_LIGHT_IN_VACUUM_2014
- STANDARD_ACCELERATION_OF_GRAVITY_2014
- STANDARD_ATMOSPHERE_2014
- STANDARD_STATE_PRESSURE_2014
- STEFAN_BOLTZMANN_CONSTANT_2014
- TAU_COMPTON_WAVELENGTH_2014
- TAU_COMPTON_WAVELENGTH_OVER_2_PI_2014
- TAU_ELECTRON_MASS_RATIO_2014
- TAU_MASS_2014
- TAU_MASS_ENERGY_EQUIVALENT_2014
- TAU_MASS_ENERGY_EQUIVALENT_IN_MEV_2014

- TAU_MASS_IN_U_2014
- TAU_MOLAR_MASS_2014
- TAU_MUON_MASS_RATIO_2014
- TAU_NEUTRON_MASS_RATIO_2014
- TAU_PROTON_MASS_RATIO_2014
- THOMSON_CROSS_SECTION_2014
- TRITON_ELECTRON_MASS_RATIO_2014
- TRITON_G_FACTOR_2014
- TRITON_MAG_MOM_2014
- TRITON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- TRITON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- TRITON_MASS_2014
- TRITON_MASS_ENERGY_EQUIVALENT_2014
- TRITON_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- TRITON_MASS_IN_U_2014
- TRITON_MOLAR_MASS_2014
- TRITON_PROTON_MASS_RATIO_2014
- UNIFIED_ATOMIC_MASS_UNIT_2014
- VON_KLITZING_CONSTANT_2014
- WEAK_MIXING_ANGLE_2014
- WIEN_FREQUENCY_DISPLACEMENT_LAW_CONSTANT_2014
- WIEN_WAVELENGTH_DISPLACEMENT_LAW_CONSTANT_2014

CODATA 2010

List of available constants:

- LATTICE_SPACING_OF_SILICON_2010
- ALPHA_PARTICLE_ELECTRON_MASS_RATIO_2010
- ALPHA_PARTICLE_MASS_2010
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_2010
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- ALPHA_PARTICLE_MASS_IN_U_2010
- ALPHA_PARTICLE_MOLAR_MASS_2010
- ALPHA_PARTICLE_PROTON_MASS_RATIO_2010
- ANGSTROM_STAR_2010
- ATOMIC_MASS_CONSTANT_2010
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_2010
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_IN_MEV_2010
- ATOMIC_MASS_UNIT_ELECTRON_VOLT_RELATIONSHIP_2010
- ATOMIC_MASS_UNIT_HARTREE_RELATIONSHIP_2010

- ATOMIC_MASS_UNIT_HERTZ_RELATIONSHIP_2010
- ATOMIC_MASS_UNIT_INVERSE_METER_RELATIONSHIP_2010
- ATOMIC_MASS_UNIT_JOULE_RELATIONSHIP_2010
- ATOMIC_MASS_UNIT_KELVIN_RELATIONSHIP_2010
- ATOMIC_MASS_UNIT_KILOGRAM_RELATIONSHIP_2010
- ATOMIC_UNIT_OF_1ST_HYPERPOLARIZABILITY_2010
- ATOMIC_UNIT_OF_2ND_HYPERPOLARIZABILITY_2010
- ATOMIC_UNIT_OF_ACTION_2010
- ATOMIC_UNIT_OF_CHARGE_2010
- ATOMIC_UNIT_OF_CHARGE_DENSITY_2010
- ATOMIC_UNIT_OF_CURRENT_2010
- ATOMIC_UNIT_OF_ELECTRIC_DIPOLE_MOM_2010
- ATOMIC_UNIT_OF_ELECTRIC_FIELD_2010
- ATOMIC_UNIT_OF_ELECTRIC_FIELD_GRADIENT_2010
- ATOMIC_UNIT_OF_ELECTRIC_POLARIZABILITY_2010
- ATOMIC_UNIT_OF_ELECTRIC_POTENTIAL_2010
- ATOMIC_UNIT_OF_ELECTRIC_QUADRUPOLE_MOM_2010
- ATOMIC_UNIT_OF_ENERGY_2010
- ATOMIC_UNIT_OF_FORCE_2010
- ATOMIC_UNIT_OF_LENGTH_2010
- ATOMIC_UNIT_OF_MAG_DIPOLE_MOM_2010
- ATOMIC_UNIT_OF_MAG_FLUX_DENSITY_2010
- ATOMIC_UNIT_OF_MAGNETIZABILITY_2010
- ATOMIC_UNIT_OF_MASS_2010
- ATOMIC_UNIT_OF_MOMUM_2010
- ATOMIC_UNIT_OF_PERMITTIVITY_2010
- ATOMIC_UNIT_OF_TIME_2010
- ATOMIC_UNIT_OF_VELOCITY_2010
- AVOGADRO_CONSTANT_2010
- BOHR_MAGNETON_2010
- BOHR_MAGNETON_IN_EV_T_2010
- BOHR_MAGNETON_IN_HZ_T_2010
- BOHR_MAGNETON_IN_INVERSE_METERS_PER_TESLA_2010
- BOHR_MAGNETON_IN_K_T_2010
- BOHR_RADIUS_2010
- BOLTZMANN_CONSTANT_2010
- BOLTZMANN_CONSTANT_IN_EV_K_2010
- BOLTZMANN_CONSTANT_IN_HZ_K_2010

- BOLTZMANN_CONSTANT_IN_INVERSE_METERS_PER_KELVIN_2010
- CHARACTERISTIC_IMPEDANCE_OF_VACUUM_2010
- CLASSICAL_ELECTRON_RADIUS_2010
- COMPTON_WAVELENGTH_2010
- COMPTON_WAVELENGTH_OVER_2_PI_2010
- CONDUCTANCE_QUANTUM_2010
- CONVENTIONAL_VALUE_OF_JOSEPHSON_CONSTANT_2010
- CONVENTIONAL_VALUE_OF_VON_KLITZING_CONSTANT_2010
- CU_X_UNIT_2010
- DEUTERON_ELECTRON_MAG_MOM_RATIO_2010
- DEUTERON_ELECTRON_MASS_RATIO_2010
- DEUTERON_G_FACTOR_2010
- DEUTERON_MAG_MOM_2010
- DEUTERON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- DEUTERON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- DEUTERON_MASS_2010
- DEUTERON_MASS_ENERGY_EQUIVALENT_2010
- DEUTERON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- DEUTERON_MASS_IN_U_2010
- DEUTERON_MOLAR_MASS_2010
- DEUTERON_NEUTRON_MAG_MOM_RATIO_2010
- DEUTERON_PROTON_MAG_MOM_RATIO_2010
- DEUTERON_PROTON_MASS_RATIO_2010
- DEUTERON_RMS_CHARGE_RADIUS_2010
- ELECTRIC_CONSTANT_2010
- ELECTRON_CHARGE_TO_MASS_QUOTIENT_2010
- ELECTRON_DEUTERON_MAG_MOM_RATIO_2010
- ELECTRON_DEUTERON_MASS_RATIO_2010
- ELECTRON_G_FACTOR_2010
- ELECTRON_GYROMAG_RATIO_2010
- ELECTRON_GYROMAG_RATIO_OVER_2_PI_2010
- ELECTRON_HELION_MASS_RATIO_2010
- ELECTRON_MAG_MOM_2010
- ELECTRON_MAG_MOM_ANOMALY_2010
- ELECTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- ELECTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- ELECTRON_MASS_2010
- ELECTRON_MASS_ENERGY_EQUIVALENT_2010

- ELECTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- ELECTRON_MASS_IN_U_2010
- ELECTRON_MOLAR_MASS_2010
- ELECTRON_MUON_MAG_MOM_RATIO_2010
- ELECTRON_MUON_MASS_RATIO_2010
- ELECTRON_NEUTRON_MAG_MOM_RATIO_2010
- ELECTRON_NEUTRON_MASS_RATIO_2010
- ELECTRON_PROTON_MAG_MOM_RATIO_2010
- ELECTRON_PROTON_MASS_RATIO_2010
- ELECTRON_TAU_MASS_RATIO_2010
- ELECTRON_TO_ALPHA_PARTICLE_MASS_RATIO_2010
- ELECTRON_TO_SHIELDED_HELION_MAG_MOM_RATIO_2010
- ELECTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2010
- ELECTRON_TRITON_MASS_RATIO_2010
- ELECTRON_VOLT_2010
- ELECTRON_VOLT_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
- ELECTRON_VOLT_HARTREE_RELATIONSHIP_2010
- ELECTRON_VOLT_HERTZ_RELATIONSHIP_2010
- ELECTRON_VOLT_INVERSE_METER_RELATIONSHIP_2010
- ELECTRON_VOLT_JOULE_RELATIONSHIP_2010
- ELECTRON_VOLT_KELVIN_RELATIONSHIP_2010
- ELECTRON_VOLT_KILOGRAM_RELATIONSHIP_2010
- ELEMENTARY_CHARGE_2010
- ELEMENTARY_CHARGE_OVER_H_2010
- FARADAY_CONSTANT_2010
- FARADAY_CONSTANT_FOR_CONVENTIONAL_ELECTRIC_CURRENT_2010
- FERMI_COUPLING_CONSTANT_2010
- FINE_STRUCTURE_CONSTANT_2010
- FIRST_RADIATION_CONSTANT_2010
- FIRST_RADIATION_CONSTANT_FOR_SPECTRAL_RADIANCE_2010
- HARTREE_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
- HARTREE_ELECTRON_VOLT_RELATIONSHIP_2010
- HARTREE_ENERGY_2010
- HARTREE_ENERGY_IN_EV_2010
- HARTREE_HERTZ_RELATIONSHIP_2010
- HARTREE_INVERSE_METER_RELATIONSHIP_2010
- HARTREE_JOULE_RELATIONSHIP_2010
- HARTREE_KELVIN_RELATIONSHIP_2010

- HARTREE_KILOGRAM_RELATIONSHIP_2010
- HELION_ELECTRON_MASS_RATIO_2010
- HELION_G_FACTOR_2010
- HELION_MAG_MOM_2010
- HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- HELION_MASS_2010
- HELION_MASS_ENERGY_EQUIVALENT_2010
- HELION_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- HELION_MASS_IN_U_2010
- HELION_MOLAR_MASS_2010
- HELION_PROTON_MASS_RATIO_2010
- HERTZ_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
- HERTZ_ELECTRON_VOLT_RELATIONSHIP_2010
- HERTZ_HARTREE_RELATIONSHIP_2010
- HERTZ_INVERSE_METER_RELATIONSHIP_2010
- HERTZ_JOULE_RELATIONSHIP_2010
- HERTZ_KELVIN_RELATIONSHIP_2010
- HERTZ_KILOGRAM_RELATIONSHIP_2010
- INVERSE_FINE_STRUCTURE_CONSTANT_2010
- INVERSE_METER_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
- INVERSE_METER_ELECTRON_VOLT_RELATIONSHIP_2010
- INVERSE_METER_HARTREE_RELATIONSHIP_2010
- INVERSE_METER_HERTZ_RELATIONSHIP_2010
- INVERSE_METER_JOULE_RELATIONSHIP_2010
- INVERSE_METER_KELVIN_RELATIONSHIP_2010
- INVERSE_METER_KILOGRAM_RELATIONSHIP_2010
- INVERSE_OF_CONDUCTANCE_QUANTUM_2010
- JOSEPHSON_CONSTANT_2010
- JOULE_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
- JOULE_ELECTRON_VOLT_RELATIONSHIP_2010
- JOULE_HARTREE_RELATIONSHIP_2010
- JOULE_HERTZ_RELATIONSHIP_2010
- JOULE_INVERSE_METER_RELATIONSHIP_2010
- JOULE_KELVIN_RELATIONSHIP_2010
- JOULE_KILOGRAM_RELATIONSHIP_2010
- KELVIN_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
- KELVIN_ELECTRON_VOLT_RELATIONSHIP_2010

- KELVIN_HARTREE_RELATIONSHIP_2010
- KELVIN_HERTZ_RELATIONSHIP_2010
- KELVIN_INVERSE_METER_RELATIONSHIP_2010
- KELVIN_JOULE_RELATIONSHIP_2010
- KELVIN_KILOGRAM_RELATIONSHIP_2010
- KILOGRAM_ATOMIC_MASS_UNIT_RELATIONSHIP_2010
- KILOGRAM_ELECTRON_VOLT_RELATIONSHIP_2010
- KILOGRAM_HARTREE_RELATIONSHIP_2010
- KILOGRAM_HERTZ_RELATIONSHIP_2010
- KILOGRAM_INVERSE_METER_RELATIONSHIP_2010
- KILOGRAM_JOULE_RELATIONSHIP_2010
- KILOGRAM_KELVIN_RELATIONSHIP_2010
- LATTICE_PARAMETER_OF_SILICON_2010
- LOSCHMIDT_CONSTANT_273_15_K_100_KPA_2010
- LOSCHMIDT_CONSTANT_273_15_K_101_325_KPA_2010
- MAG_CONSTANT_2010
- MAG_FLUX_QUANTUM_2010
- MOLAR_GAS_CONSTANT_2010
- MOLAR_MASS_CONSTANT_2010
- MOLAR_MASS_OF_CARBON_12_2010
- MOLAR_PLANCK_CONSTANT_2010
- MOLAR_PLANCK_CONSTANT_TIMES_C_2010
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_100_KPA_2010
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_101_325_KPA_2010
- MOLAR_VOLUME_OF_SILICON_2010
- MO_X_UNIT_2010
- MUON_COMPTON_WAVELENGTH_2010
- MUON_COMPTON_WAVELENGTH_OVER_2_PI_2010
- MUON_ELECTRON_MASS_RATIO_2010
- MUON_G_FACTOR_2010
- MUON_MAG_MOM_2010
- MUON_MAG_MOM_ANOMALY_2010
- MUON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- MUON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- MUON_MASS_2010
- MUON_MASS_ENERGY_EQUIVALENT_2010
- MUON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- MUON_MASS_IN_U_2010

- MUON_MOLAR_MASS_2010
- MUON_NEUTRON_MASS_RATIO_2010
- MUON_PROTON_MAG_MOM_RATIO_2010
- MUON_PROTON_MASS_RATIO_2010
- MUON_TAU_MASS_RATIO_2010
- NATURAL_UNIT_OF_ACTION_2010
- NATURAL_UNIT_OF_ACTION_IN_EV_S_2010
- NATURAL_UNIT_OF_ENERGY_2010
- NATURAL_UNIT_OF_ENERGY_IN_MEV_2010
- NATURAL_UNIT_OF_LENGTH_2010
- NATURAL_UNIT_OF_MASS_2010
- NATURAL_UNIT_OF_MOMUM_2010
- NATURAL_UNIT_OF_MOMUM_IN_MEV_C_2010
- NATURAL_UNIT_OF_TIME_2010
- NATURAL_UNIT_OF_VELOCITY_2010
- NEUTRON_COMPTON_WAVELENGTH_2010
- NEUTRON_COMPTON_WAVELENGTH_OVER_2_PI_2010
- NEUTRON_ELECTRON_MAG_MOM_RATIO_2010
- NEUTRON_ELECTRON_MASS_RATIO_2010
- NEUTRON_G_FACTOR_2010
- NEUTRON_GYROMAG_RATIO_2010
- NEUTRON_GYROMAG_RATIO_OVER_2_PI_2010
- NEUTRON_MAG_MOM_2010
- NEUTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- NEUTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- NEUTRON_MASS_2010
- NEUTRON_MASS_ENERGY_EQUIVALENT_2010
- NEUTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- NEUTRON_MASS_IN_U_2010
- NEUTRON_MOLAR_MASS_2010
- NEUTRON_MUON_MASS_RATIO_2010
- NEUTRON_PROTON_MAG_MOM_RATIO_2010
- NEUTRON_PROTON_MASS_DIFFERENCE_2010
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_2010
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_IN_MEV_2010
- NEUTRON_PROTON_MASS_DIFFERENCE_IN_U_2010
- NEUTRON_PROTON_MASS_RATIO_2010
- NEUTRON_TAU_MASS_RATIO_2010

- NEUTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2010
- NEWTONIAN_CONSTANT_OF_GRAVITATION_2010
- NEWTONIAN_CONSTANT_OF_GRAVITATION_OVER_H_BAR_C_2010
- NUCLEAR_MAGNETON_2010
- NUCLEAR_MAGNETON_IN_EV_T_2010
- NUCLEAR_MAGNETON_IN_INVERSE_METERS_PER_TESLA_2010
- NUCLEAR_MAGNETON_IN_K_T_2010
- NUCLEAR_MAGNETON_IN_MHZ_T_2010
- PLANCK_CONSTANT_2010
- PLANCK_CONSTANT_IN_EV_S_2010
- PLANCK_CONSTANT_OVER_2_PI_2010
- PLANCK_CONSTANT_OVER_2_PI_IN_EV_S_2010
- PLANCK_CONSTANT_OVER_2_PI_TIMES_C_IN_MEV_FM_2010
- PLANCK_LENGTH_2010
- PLANCK_MASS_2010
- PLANCK_MASS_ENERGY_EQUIVALENT_IN_GEV_2010
- PLANCK_TEMPERATURE_2010
- PLANCK_TIME_2010
- PROTON_CHARGE_TO_MASS_QUOTIENT_2010
- PROTON_COMPTON_WAVELENGTH_2010
- PROTON_COMPTON_WAVELENGTH_OVER_2_PI_2010
- PROTON_ELECTRON_MASS_RATIO_2010
- PROTON_G_FACTOR_2010
- PROTON_GYROMAG_RATIO_2010
- PROTON_GYROMAG_RATIO_OVER_2_PI_2010
- PROTON_MAG_MOM_2010
- PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- PROTON_MAG_SHIELDING_CORRECTION_2010
- PROTON_MASS_2010
- PROTON_MASS_ENERGY_EQUIVALENT_2010
- PROTON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- PROTON_MASS_IN_U_2010
- PROTON_MOLAR_MASS_2010
- PROTON_MUON_MASS_RATIO_2010
- PROTON_NEUTRON_MAG_MOM_RATIO_2010
- PROTON_NEUTRON_MASS_RATIO_2010
- PROTON_RMS_CHARGE_RADIUS_2010

- PROTON_TAU_MASS_RATIO_2010
- QUANTUM_OF_CIRCULATION_2010
- QUANTUM_OF_CIRCULATION_TIMES_2_2010
- RYDBERG_CONSTANT_2010
- RYDBERG_CONSTANT_TIMES_C_IN_HZ_2010
- RYDBERG_CONSTANT_TIMES_HC_IN_EV_2010
- RYDBERG_CONSTANT_TIMES_HC_IN_J_2010
- SACKUR_TETRODE_CONSTANT_1_K_100_KPA_2010
- SACKUR_TETRODE_CONSTANT_1_K_101_325_KPA_2010
- SECOND_RADIATION_CONSTANT_2010
- SHIELDED_HELION_GYROMAG_RATIO_2010
- SHIELDED_HELION_GYROMAG_RATIO_OVER_2_PI_2010
- SHIELDED_HELION_MAG_MOM_2010
- SHIELDED_HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- SHIELDED_HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- SHIELDED_HELION_TO_PROTON_MAG_MOM_RATIO_2010
- SHIELDED_HELION_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2010
- SHIELDED_PROTON_GYROMAG_RATIO_2010
- SHIELDED_PROTON_GYROMAG_RATIO_OVER_2_PI_2010
- SHIELDED_PROTON_MAG_MOM_2010
- SHIELDED_PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- SHIELDED_PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- SPEED_OF_LIGHT_IN_VACUUM_2010
- STANDARD_ACCELERATION_OF_GRAVITY_2010
- STANDARD_ATMOSPHERE_2010
- STANDARD_STATE_PRESSURE_2010
- STEFAN_BOLTZMANN_CONSTANT_2010
- TAU_COMPTON_WAVELENGTH_2010
- TAU_COMPTON_WAVELENGTH_OVER_2_PI_2010
- TAU_ELECTRON_MASS_RATIO_2010
- TAU_MASS_2010
- TAU_MASS_ENERGY_EQUIVALENT_2010
- TAU_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- TAU_MASS_IN_U_2010
- TAU_MOLAR_MASS_2010
- TAU_MUON_MASS_RATIO_2010
- TAU_NEUTRON_MASS_RATIO_2010
- TAU_PROTON_MASS_RATIO_2010

- THOMSON_CROSS_SECTION_2010
- TRITON_ELECTRON_MASS_RATIO_2010
- TRITON_G_FACTOR_2010
- TRITON_MAG_MOM_2010
- TRITON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- TRITON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- TRITON_MASS_2010
- TRITON_MASS_ENERGY_EQUIVALENT_2010
- TRITON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- TRITON_MASS_IN_U_2010
- TRITON_MOLAR_MASS_2010
- TRITON_PROTON_MASS_RATIO_2010
- UNIFIED_ATOMIC_MASS_UNIT_2010
- VON_KLITZING_CONSTANT_2010
- WEAK_MIXING_ANGLE_2010
- WIEN_FREQUENCY_DISPLACEMENT_LAW_CONSTANT_2010
- WIEN_WAVELENGTH_DISPLACEMENT_LAW_CONSTANT_2010

NAME

codata - fundamental physical constants

SYNOPSIS

codata [OPTION...] [REGEX_PATTERN...]

DESCRIPTION

codata(1) is a command line interface which prints all the **codata** constants.

The current values are from 2022. Older values can be retrieved if needed and the output can be filtered with REGEX PATTERNS.

OPTIONS

--year, -y YEAR
: Codata constants: 2022, 2018, 2014, 2010

--pattern, -p PATTERN
: Regex pattern for filtering the constants.

--value, -a
: Show only the value.

--error, -e
: Show only the uncertainty.

--usage
: Show usage text and exit.

--version, -v
: Show version information and exit.

--help, -h
: Show help text and exit.

NOTES

You may replace the default options from a file if your first options begin with @file. Initial options will then be read from the "response file" "file.rsp" in the current directory.

If "file" does not exist or cannot be read, then an error occurs and the program stops. Each line of the file is prefixed with "options" and interpreted as a separate argument. The file itself may not contain @file arguments. That is, it is not processed recursively.

For more information on response files see https://urbanjost.github.io/M_CLI2/set_args.3m_cli2.html

EXAMPLE

Minimal example

```
codata
codata -y 2018 molar electron
codata -y 2014 -p molar.*gas,electron.*eV
codata [B,b]oltzmann.*eV
```

SEE ALSO

codata(3)

NAME

codata_constant_type - data type defining a constant

SYNOPSIS

```
Fortran: type(codata_constant_type) :: c

C: struct codata_constant_type c
```

DESCRIPTION

Defines a constant with the name, the value, the uncertainty and the unit.

EXAMPLE

Fortran:

```
use codata
type(codata_constant_type), pointer :: c
c = SPEED_OF_LIGHT_IN_VACUUM
```

C:

```
struct codata_constant_type c;
c = SPEED_OF_LIGHT_IN_VACUUM;
```

SEE ALSO

codata(3)

NAME

version - get the version of the Fortran library codata

LIBRARY

codata (**-libcodata**, **-lcodata**)

SYNOPSIS

```
Fortran: character(len=:), pointer version()

C: char* codata_version()

Python: pycodata.__version__
```

DESCRIPTION

Get the library version.

EXAMPLE

Fortran:

```
use codata
print *, "version = ", version()
```

C:

```
include <stdio.h>
include "codata.h"
printf("version = %s0, codata_version());
```

Python:

```
import pycodata
print(f"version = {pycodata.__version__}")
```

SEE ALSO

codata(3)